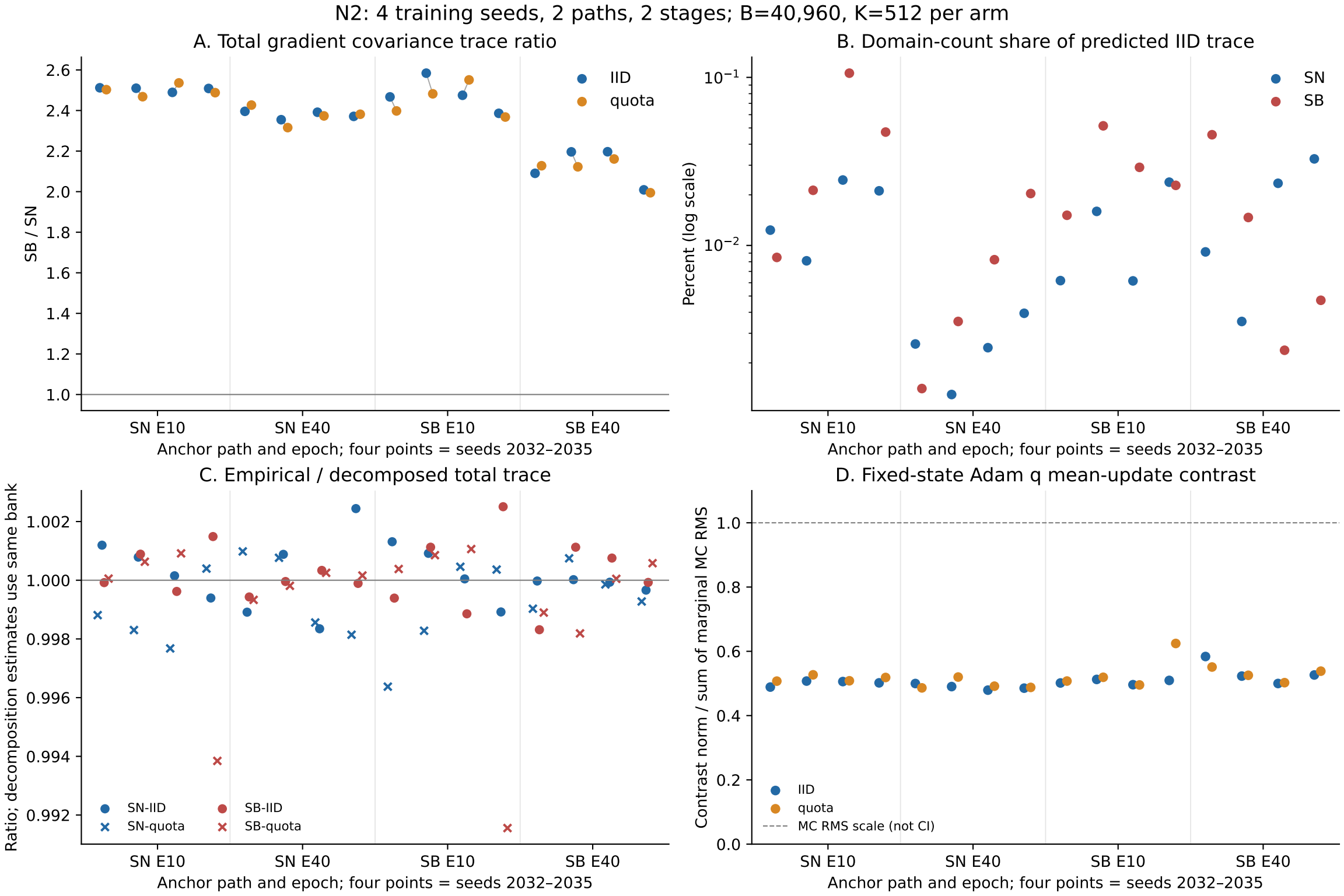
N2: 4 training seeds, 2 paths, 2 stages; B=40,960, K=512 per arm
A. Total gradient covariance trace ratio
SB / SN
IID
quota
SN E10
SN E40
SB E10
SB E40
Anchor path and epoch; four points = seeds 2032–2035
B. Domain-count share of predicted IID trace
Percent (log scale)
SN
SB
SN E10
SN E40
SB E10
SB E40
Anchor path and epoch; four points = seeds 2032–2035
C. Empirical / decomposed total trace
Ratio; decomposition estimates use same bank
SN-IID
SN-quota
SB-IID
SB-quota
SN E10
SN E40
SB E10
SB E40
Anchor path and epoch; four points = seeds 2032–2035
D. Fixed-state Adam q mean-update contrast
Contrast norm / sum of marginal MC RMS
IID
quota
MC RMS scale (not CI)
SN E10
SN E40
SB E10
SB E40
Anchor path and epoch; four points = seeds 2032–2035